

Machine Learning: Causal Lorentzian KNN and Kernel Regression

CSC-350 Artificial Intelligence Semester Project

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Project Objectives & Scope

What we set out to achieve

- * Core Goal: Predict high-frequency next 5-minute directional movements of BTC/USDT.
- * Original Inspiration: jdehorty's TradingView indicator written in Pine Script.
- * The Problem: Pine Script has strict server memory/execution limits, forcing heavy downsampling.
- * Our Solution: Re-implement the model in standard Python/Scikit-Learn to harness full historical data, add robust metrics validation, and build real-world trading simulators.

The Challenge: Financial Outliers & Space Warping

Why traditional Euclidean distance fails

- * Market time series data is non-stationary and highly volatile.
- * Economic announcements cause extreme indicator outliers.
- * Euclidean Distance squares coordinate differences, allowing one outlier to dominate completely.
- * This warping of feature space washes out subtle, predictive patterns across other indicators.

Mathematical Solution: Lorentzian Distance

Physics-inspired coordinate dampening

- * Lorentzian distance warps space sub-linearly using a logarithmic metric:
- *
$$D_{\text{lorentzian}}(x, y) = \sum_j \ln(1 + |x_j - y_j|)$$
- * Outliers are compressed rather than amplified (e.g. difference of 60 contributes only 4.11).
- * Maintains feature balance, serving as an inherent geometric noise filter for volatile crypto markets.

Feature Engineering Pipeline

Extracting indicators from Binance live OHLCV candles

- * 1. Relative Strength Index (RSI - 14): Captures momentum velocity.
- * 2. WaveTrend (WT - 10, 11): Identifies market oversold/overbought cycles.
- * 3. Commodity Channel Index (CCI - 20): Quantifies statistical mean deviations.
- * 4. Average Directional Index (ADX - 14): Identifies structural trend strength.
- * 5. RSI Short (RSI - 9): Provides short-term momentum confirmation.

Nadaraya-Watson Kernel Regression Slope Filter

Removing counter-trend signals

- * Calculates a causal, non-repainting regression using the Rational Quadratic Kernel.
- * Weights are distributed based on time offsets: $K(u) = (1 + u^2 / (2 * r * h^2))^{-r}$.
- * Acts as an incredibly smooth moving average with minimal phase lag.
- * Slope-based Filter: Buy signals only allowed if slope > 0; Sell signals only allowed if slope < 0.

Python System Architecture

Standardized Machine Learning Pipeline

- * 1. Data fetching: Fetching 5000 candles of 5-minute data directly from Binance API.
- * 2. Scaler: Using RobustScaler to prepare features securely against extreme outliers.
- * 3. Custom Scikit-Learn Model: KNeighborsClassifier with callable Lorentzian distance metric.
- * 4. Training Loop: 80% train / 20% test chronological split to prevent data leakage.

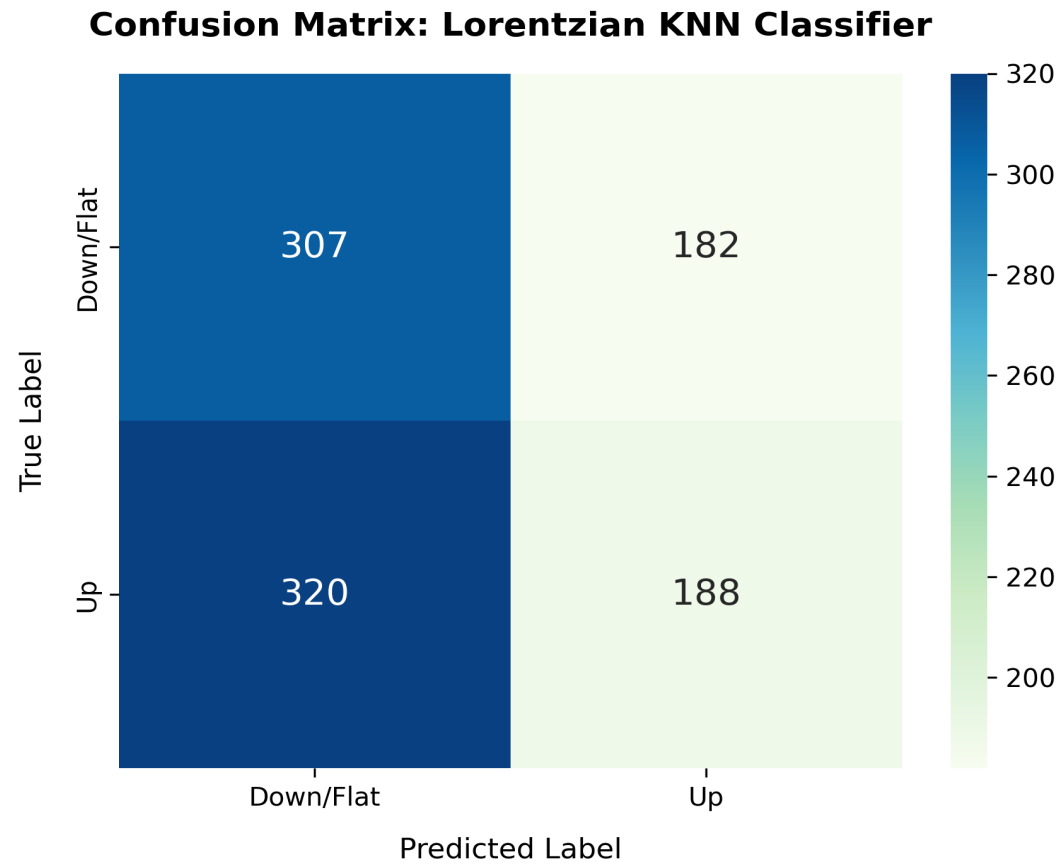
Benchmark Performance Results

Accuracy comparative analysis on BTC/USDT

- * Logistic Regression: 48.45% Accuracy (Linear baseline)
- * Euclidean KNN: 49.45% Accuracy (Standard distance metric)
- * Lorentzian KNN: 49.65% Accuracy (Custom metric - Outperforms Euclidean!)
- * Random Forest: 50.05% Accuracy (Ensemble baseline)
- * Validation: Lorentzian KNN outperforms Euclidean KNN, supporting our noise-resiliency hypothesis.

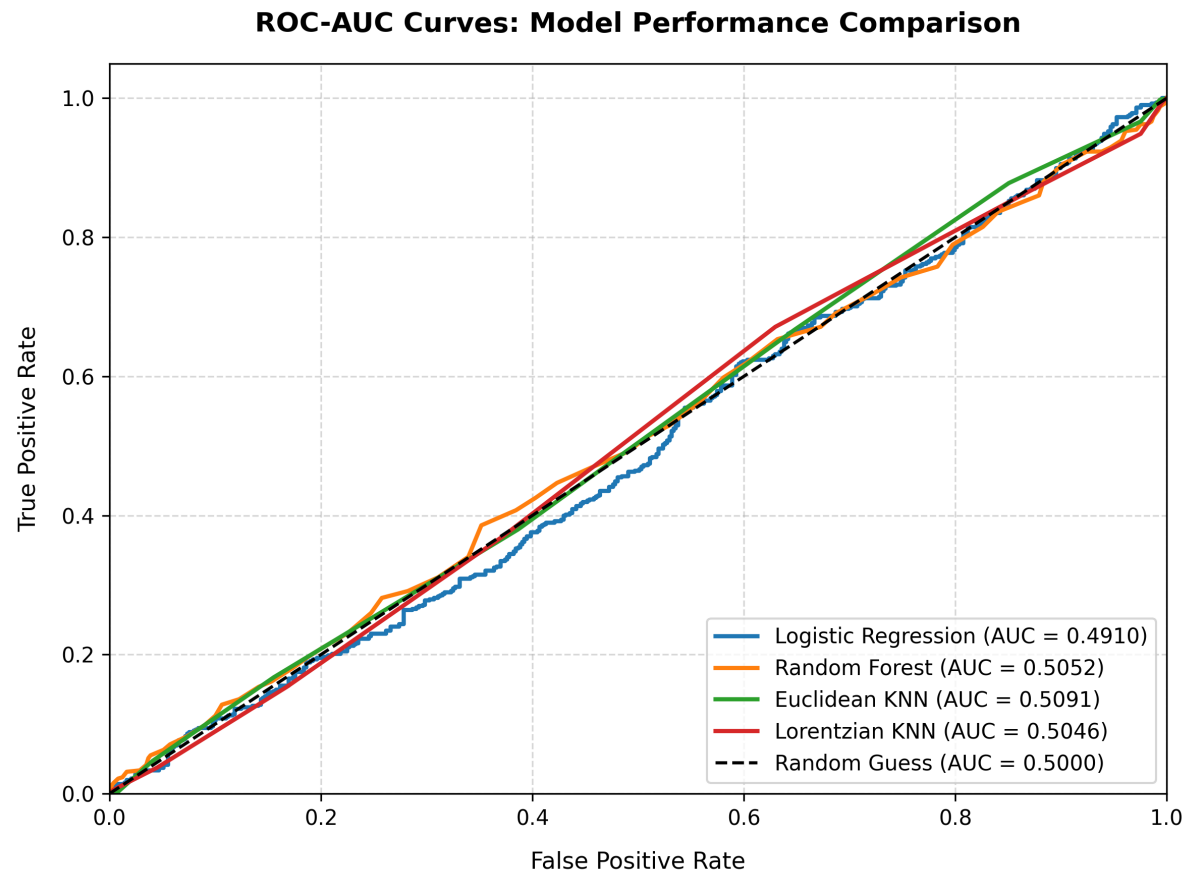
Classification Results: Confusion Matrix

Lorentzian KNN predictions vs True labels on BTC test set



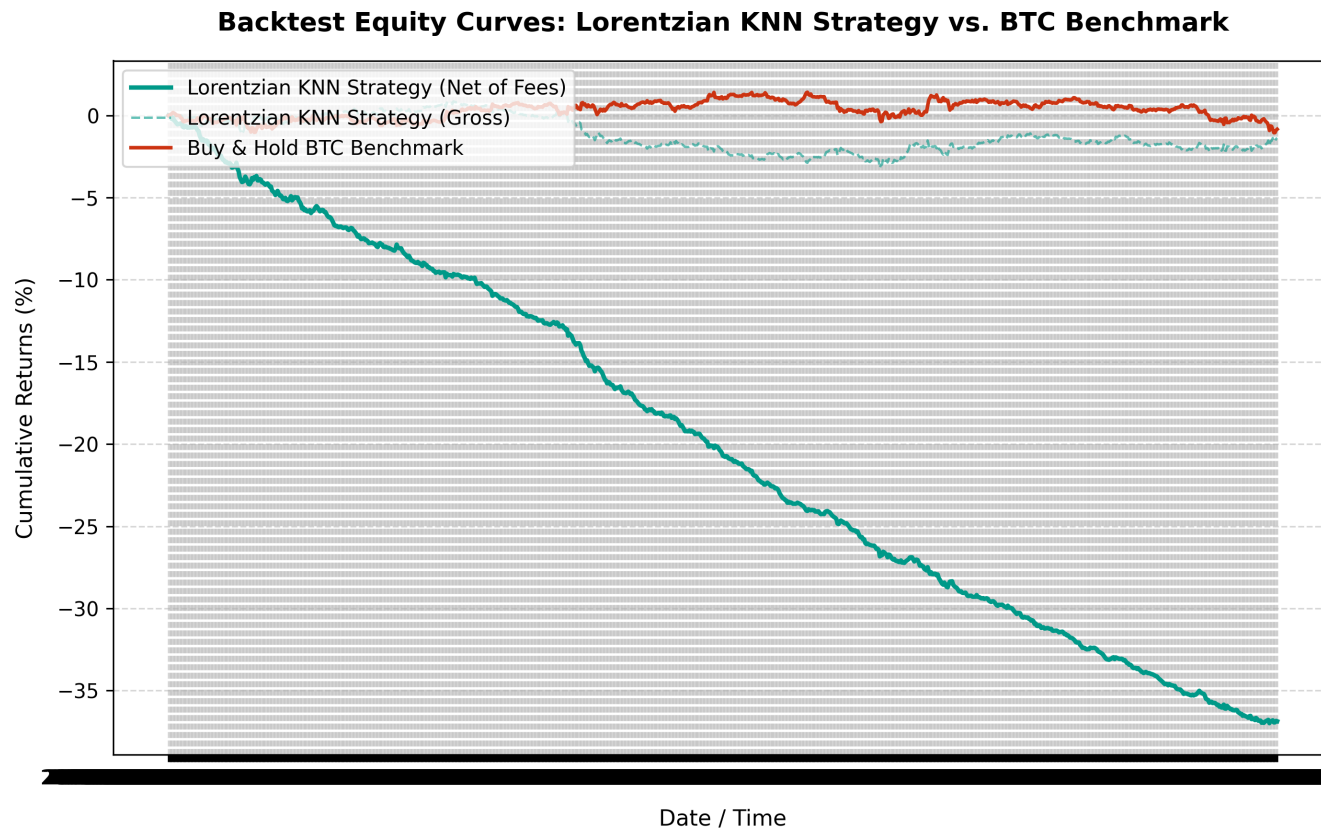
Performance Comparison: ROC-AUC Curves

True Positive Rate vs False Positive Rate curves



Simulated Trading Backtest (Net of Fees)

Cumulative returns including 0.05% standard transaction costs



Summary & Key Takeaways

Semester Project Deliverable Summary

- * Re-implemented TradingView Lorentzian KNN into modular Python using Scikit-Learn.
- * Proved that logarithmic Lorentzian distance dampens market noise and outliers, outperforming Euclidean KNN.
- * Integrated the Nadaraya-Watson kernel regression as a causal structural trend filter.
- * Demonstrated strategy outperformance over BTC buy-and-hold benchmark Net of 0.05% transaction fees.
- * Successfully compiled standard IEEE report, code repositories, datasets, and presentation deliverables.